

Youshi (Tiger) Chen

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EDUCATION

Stanford University Stanford, CA
B.S., Mathematics and Computer Science Expected 2030
Fall 2026: MATH 51 (Linear Algebra & Multivariable Calculus), CS 106B (Programming Abstractions), PHYSICS 41 (Mechanics)
Troy High School, Troy Tech Magnet Program Fullerton, CA
High School Diploma, GPA 4.84 | SAT 1540 2022 - 2026
Coursework: Multivariable Calculus, Differential Equations & Linear Algebra, AP Physics C, AP Computer Science A

PROJECTS

Options Dealer-Positioning & 0DTE Strategy Research (*Python, 280+ commits*) 2026 - Present

- Built a Python tool computing dealer gamma (GEX), vanna, and charm exposure, zero-gamma flip levels, and call/put walls from option open interest via Black-Scholes Greeks, with heatmap dashboards and live alerts on a cloud VM.
- Backtested GEX-gated 0DTE SPY breakout strategies over 268 trading days (3,390 entries) on real NBBO option quotes against matched-null controls; found no tradeable edge and killed the strategy rather than tune it.
- Caught two lookahead bugs in self-audits (a 4-minute confirmation peek and an exit-fill convention) that had inflated a 36% win rate to 63% and flipped -5.5% per trade to +1.7%; live paper-trading fills confirmed the corrected result (31% win rate, $t = -2.74$).
- Built a 4-stage strategy validator (deflated Sharpe, cross-instrument replication, embargoed walk-forward, sealed holdout) that rejected all 202 hypotheses tested.
- Tested SPY mean-reversion against a rolling-OLS cross-market fair value (QQQ, VIX, rates, breadth) across a pre-registered 54-configuration grid on 10 years of data; concluded the apparent signal was multiple-testing noise.

Intraday Volume-Anomaly Signal for SPY/QQQ (*Pine Script v6*) 2026

- Built a 1-minute trading signal that flags six Wyckoff volume patterns by scoring each bar against trailing baselines (z-score).
- Eliminated lookahead bias with lagged statistics, close-confirmed signals, and a next-bar-open fill rule; fixed a pre-market volume artifact that fired false signals at the open.

Badminton Racket IMU Data Logger (*ESP32, KiCad, C++*) 2025

- Designed a custom ESP32 PCB with an onboard IMU that samples 3-axis swing acceleration and impact force at 1,000 Hz, producing time-series data for quantitative analysis of stroke mechanics.

EXPERIENCE

Aircart Brea, CA
Co-Founder, Autonomous Drone Engineering Lead 2025 - 2026

- Led a 4-person team from schematic to flight-ready UAV in 2 months; implemented and tuned PID flight-control loops from flight-log data across 20+ test flights to stabilize the airframe at a 5 kg payload.

FRC Robotics Fullerton, CA
Board Member, Electrical and Mechanical Team 2024 - 2026

- Designed and validated sensor networks and power distribution for a 100+ lb competition robot, using structured diagnostics to isolate and eliminate faults under match-day time pressure.

Atlas Assembly / Glenair Arcadia, CA
Engineering Intern Jun 2025 - Jul 2025

- Fabricated assembly tooling in SOLIDWORKS and validated prototypes with engineers to aerospace production standards.

COMPETITIVE ATHLETICS

Team USA Badminton, Yonex-Sponsored Athlete 2014 - Present

- Gold medalist, 2025 Pan Am Junior Championships; ranked #73 in the world; 12 years of decision-making under pressure.

SKILLS & TRADING

Programming: Python (pandas, NumPy), C++, Java, Pine Script v6, Git, Linux/systemd
Quantitative: options Greeks (gamma, vanna, charm), Black-Scholes, backtesting, lookahead-bias control, walk-forward validation, deflated Sharpe, signal processing, PID control
Hardware: KiCad, PCB layout, ESP32, IMUs, embedded data acquisition, SOLIDWORKS (CSWA, 2026), Fusion 360
Trading: Topstep funded futures trader (active); passed the Trading Combine and manage the account under a hard daily loss limit and trailing maximum-drawdown rules, sizing risk per trade to stay within them